

Working Mechanisms of Physical and MAC Layers from 1G to 6G

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1 Introduction

The OSI reference model breaks down network functions into layered abstractions. For mobile wireless systems, two critical layers are the Physical layer and the Medium Access Control sublayer. The Physical layer encompasses the raw transmission of bits over a communication channel, addressing modulation, channel coding, and signal propagation aspects. The MAC layer, part of the Data Link layer, orchestrates how multiple users access the shared spectrum, controlling when and how devices transmit data to prevent collisions, ensure fairness, and optimize throughput.

Over successive generations of mobile technology—1G (analog) to 6G (envisioned)—the underlying PHY and MAC mechanisms have evolved dramatically. These changes have been driven by the need for higher data rates, lower latencies, better coverage, energy efficiency, and support for massive device connectivity.

2 Physical Layer Mechanisms

2.1 First Generation (1G)

Technology & Modulation: Early 1G systems, such as Advanced Mobile Phone System (AMPS), relied on analog frequency modulation (FM) over large frequency allocations. The

PHY layer here was relatively simple:

- **Analog Transmission:** Continuous-wave FM signals.
- **Channelization:** Frequency Division Multiple Access (FDMA) allocated distinct frequency bands to separate voice calls.
- **Limited Data Handling:** The focus was on voice services; data capability and sophisticated error correction were minimal.

2.2 Second Generation (2G)

Key Standards: GSM, IS-95

- **Digital Modulation:** Shifted to digital techniques such as Gaussian Minimum Shift Keying (GMSK) in GSM and Code Division Multiple Access (CDMA) spreading in IS-95.
- **Forward Error Correction (FEC):** Use of convolutional codes and interleaving to improve link robustness.
- **Multiple Access:** GSM continued to rely on TDMA/FDMA hybrid, while IS-95 used CDMA for improved capacity.

The PHY layer's shift to digital modulation and robust coding improved spectrum efficiency and quality.

2.3 Third Generation (3G)

Key Standards: WCDMA (UMTS), CDMA2000

- **Wideband CDMA:** Broader bandwidth channels (5 MHz) allowed for higher data rates.
- **Advanced Coding:** Turbo codes offered significant error correction improvements.
- **Adaptive Modulation & Coding:** Networks began adjusting modulation schemes and coding rates to channel conditions, improving data throughput in variable radio environments.

2.4 Fourth Generation (4G)

Key Standards: LTE, LTE-Advanced

- **OFDM (Orthogonal Frequency Division Multiplexing):** A cornerstone modulation technique splitting the channel into numerous orthogonal subcarriers, mitigating multi-path fading and improving spectral efficiency.
- **MIMO (Multiple Input, Multiple Output):** Multiple antennas at transmitter and receiver leveraged spatial diversity and multiplexing, drastically increasing data rates and reliability.
- **Advanced Channel Coding:** Turbo codes evolved, and later LTE-Advanced included low-density parity-check (LDPC) codes for higher throughput and lower latency.
- **Carrier Aggregation:** Combining multiple frequency blocks to increase bandwidth and achieve higher peak rates.

2.5 Fifth Generation (5G)

Key Standards: 5G NR (New Radio)

- **Flexible Numerology & OFDM Variants:** 5G NR uses scalable subcarrier spacing and multiple OFDM-based waveforms to adapt to different frequency bands and use cases (eMBB, URLLC, mMTC).
- **Advanced MIMO & Massive MIMO:** Large antenna arrays (64 or more antenna elements) provide beamforming, spatial multiplexing, and improved coverage.
- **Polar & LDPC Codes:** Enhanced channel coding tailored for both control channels (Polar codes) and data channels (LDPC) to meet stringent throughput and reliability demands.
- **mmWave & Sub-THz Frequencies:** The PHY supports operation in millimeter-wave bands for ultra-high bandwidth, though with shorter coverage ranges.

2.6 Sixth Generation (6G) – Emerging Concepts

Though 6G standards are still under research and conceptualization, the PHY layer is expected to incorporate:

- **THz Band Communications:** Even higher frequency operations for unprecedented bandwidth and data rates.
- **Advanced Waveforms & Modulations:** New modulation schemes resilient to extremely high Doppler shifts and massive device densities.
- **AI-Enhanced Signal Processing:** Adaptive, intelligent PHY layer mechanisms using machine learning for channel estimation, beam selection, and interference management.
- **Quantum & Photonic Technologies:** Early explorations into quantum communication methods and photonic integrated circuits could revolutionize the PHY layer.

3 MAC Layer Mechanisms

The MAC layer ensures fair and efficient sharing of the wireless medium. Its design must consider aspects such as collision avoidance, resource allocation, and adapting to varying network conditions.

3.1 First Generation (1G)

- **Fixed Allocation Schemes:** FDMA assigned fixed frequency channels to users, and the MAC-like functionalities were rudimentary. There was minimal intelligence in scheduling and no dynamic resource allocation for data since services were predominantly voice.

3.2 Second Generation (2G)

- **TDMA/FDMA (GSM):** Time slots in a frame structure provided scheduled access. Each user received a fixed time slot, leading to predictable but not flexible resource usage.

- **CDMA (IS-95):** Users shared the same spectrum distinguished by unique spreading codes. The MAC involved power control to manage interference and maintain signal quality, rather than explicit time-slot or frequency allocation.

3.3 Third Generation (3G)

- **Capacity on Demand:** WCDMA utilized a soft capacity concept. While the MAC mechanisms were still relatively static, base stations began dynamically adjusting resources (power and spreading codes) to meet varying demand.
- **QoS Differentiation:** MAC protocols started to incorporate Quality of Service (QoS) frameworks, prioritizing certain types of traffic (voice vs. data) when resource conflicts arose.

3.4 Fourth Generation (4G)

- **Scheduler-Based MAC:** LTE's MAC layer employed a centralized scheduler at the base station (eNodeB) to dynamically allocate frequency-time resources (Resource Blocks) to users based on channel quality indicators (CQI), QoS requirements, and fairness constraints.
- **HARQ (Hybrid Automatic Repeat Request):** The MAC included mechanisms for rapid retransmission of erroneous data packets with incremental redundancy.
- **Adaptive Resource Allocation:** The MAC dynamically balanced resources among users, improving efficiency and overall system throughput.

3.5 Fifth Generation (5G)

- **Dynamic and Flexible Scheduling:** The gNodeB (5G base station) manages resources over a broader set of numerologies and frequency bands, supporting diverse traffic types: Enhanced Mobile Broadband (eMBB), Ultra-Reliable Low-Latency Communications (URLLC), and massive Machine-Type Communications (mMTC).

- **Slicing and Isolation:** MAC plays a role in network slicing, ensuring that virtual network segments get allocated resources according to their service level agreements.
- **Grant-Free Access:** For some IoT and low-latency scenarios, 5G supports grant-free uplink transmissions, reducing latency and overhead by letting certain devices transmit without waiting for a scheduling grant.

3.6 Sixth Generation (6G) – Emerging Concepts

- **AI-Driven MAC:** Intelligent MAC scheduling may rely on machine learning to predict traffic, user movement, and channel conditions, enabling proactive resource allocation.
- **Ultra-Low Latency and High Reliability:** New MAC protocols might leverage distributed ledger technologies or coded random access schemes to support massive device connectivity and near-instantaneous response times.
- **Resource Brokering Across Heterogeneous Layers:** 6G MAC may handle multi-dimensional resource allocation (time, frequency, space, code, energy) across diverse spectra and integrate seamlessly with non-terrestrial networks (satellites, drones).